

Equiper, The Plan From Here

Repositories: cirquen-labs and wairegimaina/hq_server | Written: 18 September 2026 Supersedes the sequencing in 04_PLAN_TO_9_OF_10.pdf, which covered CalSoft alone

1. Where the system stands

534 Django tests and 61 sync tests pass. Four areas have been worked on, and they are at different stages.

Area	State	Score / coverage
CalSoft , calibration and certification	Phases 1, 2, 3 and most of 5 done	8.6 / 10 , 188 tests
calSchedules , planning and grouping	Auto-scheduling repaired, regrouping added	126 tests
HOD dashboard , the head's first screen	Rebuilt around calibration coverage	14 tests, from none
audit_log , the audit trail	Both trails now surfaced	17 tests, from none
HQ server , sync and certificate authority	One of six items done (B1)	41 tests, unaudited in parts

Two things are worth saying plainly about that table.

The gains came from wiring, not from new mathematics. Almost every fix was a correct component that nothing called, or two components that disagreed: guard banding written and never invoked, a drift engine whose consumer read keys it never returned, two definitions of group membership, two certificate allocators, an audit trail written from five places and displayed nowhere. The system's problem has not been capability. It has been that capable parts were not connected to each other.

Three apps had no tests at all before this work, and two of them (dashboard, audit_log) were changed during it. Writing the tests alongside the changes is the only reason those changes can be trusted; reporthub, machineReports and parts_tools still have none and remain the least safe places in the codebase to change.

2. What is left, by consequence

Ordered by what goes wrong if it is not done, not by effort.

Tier 1, a certificate can still be doubted

#	Item	Where	Effort
1.1	HQ refuses to null an allocated certificate number (B4)	HQ	1 d
1.2	Confirm the H-6 race empirically	HQ	0.5 d
1.3	Make the last-write-wins guard able to engage (B3)	HQ + sync	3-5 d
1.4	Void and reissue instead of reassigning (B2)	HQ + sync	3-5 d
1.5	Sign the certificate; make verification real (P4)	HQ + CalSoft	1-2 w

1.1 is still the only item that closes an *active* data-loss path. Everything else in this tier prevents a future one.

1.5 is what the certificate currently promises and does not do. The QR note and footer no longer overstate it, so the document is honest, but an issued certificate is still not tamper-evident, and that is the single largest gap against Fluke and Beamex.

Tier 2, two decisions are gating four items

Neither is a technical question, and both change what stored data means.

#	Decision	What it unblocks
2.1	Sign convention. Error of indication (mean - reference) or correction (reference - mean)?	P1.4, plus a data migration and the on-screen preview, which currently computes the opposite sign to the certificate
2.2	Minimum readings. Reject below num_readings, warn and certify, or cap k and disclose the truncation?	Enforcement, plus the $k = 12.706$ case that two readings now produce

On 2.2: capping k understates the interval, which is the dishonest direction so it was not chosen unilaterally. The recommendation is to reject below the parameter's declared num_readings, but that changes what the field may submit, so it is yours.

Tier 3, completing the metrology

#	Item	Why it is not Tier 1
3.1	Sensitivity coefficients (P5.2)	The budget cannot represent an influence quantity acting indirectly, a real limit, but no current parameter needs one
3.2	Type B components as a table (P5.3)	Follows from 3.1; do them in one migration
3.3	Store the three-valued verdict (P5.4)	The certificate reports PASS/FAIL/INDETERMINATE; the database still stores two. Needs a decision of its own: what does INDETERMINATE do to a session's overall pass?
3.4	Round once, at presentation (P3.7)	Four sequential roundings; immaterial at mmHg, not at millivolts

3.1 and 3.2 share a migration with 1.3's source_updated_at. Doing all three in one pass is less risky than three passes over the same synced tables.

Tier 4, the parts nobody has looked at

Area	Lines	Tests	Risk
reporthub	2,726	0	Generates reports leadership acts on
machineReports	3,606	0	Now linked from every workshop card
parts_tools	3,038	0	Stock and tooling
HQ og_server_core, Redis layers, soft_delete_handler	~6,000	partial	Sync correctness

machineReports moved up this list during this work: the HOD dashboard now links into it from every workshop card, so more people will reach code that nothing verifies.

3. The order to do it in

NOW	1.1 null guard (1d) closes active loss	2.1 + 2.2 decisions they gate four items
WEEK 1-2	1.2 confirm H-6 (0.5d) 1.3 LWW guard live (3-5d)	P1.4 sign convention once 2.1 is settled
WEEK 3-4	1.4 void and reissue 3.1 + 3.2 + 1.3 migration in one pass	3.4 round once
WEEK 5-7	1.5 sign the certificate needs 1.1, 1.3, 1.4 first	
ONGOING	Tier 4: a test file per module before the next change to it	

The one sequencing rule that matters: 1.5 comes after 1.1-1.4. Signing a certificate whose number might later be reallocated, or whose record a stale upload can overwrite, produces a signature that attests to nothing. The integrity of the number has to be settled before it is worth signing.

4. How to work on this system

Four practices earned their place during this work. They are written down because each one caught something that tests alone did not.

Render the thing and look at it. The certificate printed two filled squares where an emoji had no glyph in the PDF font, and the workshop card reported "5 locked" on a finished group, technically true, operationally useless. Both passed every test. Neither survived being looked at.

Pin behaviour you intend to change. Tests marked PINS CURRENT BEHAVIOUR made four later changes visible instead of silent: the drift grading, the decimal precision, the coverage factor and the approval path all broke a test on purpose, and the diff showed exactly what the change meant.

Fix the class, not the instance. `test_no_broad_handler_discards_an_exception_silently` walks every module's AST and fails if a broad handler swallows an error. That prevents the next 26 instances, which fixing 26 handlers would not.

Check the claim before repeating it. Twice in this work a confident statement was wrong: the HQ server does have tests, and a narrow `except ValueError` is good design rather than a defect. Both were corrected by looking rather than by remembering. The module's own 6,008-line handbook is stale in exactly this way, superb, and wrong about the finding it calls most important.

5. What "done" looks like

Not a score. Eight properties, of which **five now hold**:

1. **[done]** One verdict, the certificate, the record and the QR agree.
2. **[done]** A budget that reconciles, with n and the applied k on the page.
3. **[done]** Uncertainty that changes the answer, guard banding, three outcomes, TUR.
4. **[done]** Analyses that fail loudly rather than printing reassurance.

5. **[done]** Formulas that cannot silently change, one implementation, each pinned.
6. **[open]** A tamper-evident certificate, signed by the component that allocates it.
7. **[open]** Records that survive the trip, the LWW guard engaging on client uploads.
8. **[open]** A budget that can represent any influence quantity.

Items 6 and 7 are Tier 1. Item 8 is Tier 3. The three remaining properties are the difference between a system that computes correctly and one that can prove what it computed.

Effort estimates assume one engineer familiar with the codebase. Items marked HQ require access to the second repository and to its database for the migration in 1.3.

